

Economics S5040: Dynamic Stochastic Modeling

Problem Set 4 — Dynamic Programming

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Instructions

Submit one Jupyter notebook containing all derivations, Python code, figures, numerical answers, and economic explanations.

You may consult lecture notes, classmates, textbooks, and AI tools. However, your notebook must make clear that you understand every result you report. A numerical answer without an economic explanation will receive little credit.

Problem 1. The Bellman Equation and the Euler Equation

Consider the deterministic optimal-growth problem

$$\max_{\{k_{t+1}\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t \log c_t$$

subject to

$$c_t + k_{t+1} = Ak_t^\alpha, \quad 0 < \alpha < 1, \quad 0 < \beta < 1.$$

- (a) Identify the five elements of this dynamic program: state, control, reward function, transition rule, and discount factor.
- (b) Write the Bellman equation.
- (c) Derive the first-order condition with respect to k_{t+1} .
- (d) Use the envelope condition to show that the first-order condition can be written as the Euler equation

$$u'(c_t) = \beta u'(c_{t+1}) f'(k_{t+1}).$$

Explain the economic meaning of this equation in terms of an intertemporal marginal rate of substitution and the marginal return to investment.

- (e) Guess

$$V(k) = E + F \log k.$$

Show that

$$F = \frac{\alpha}{1 - \alpha\beta}$$

and derive the optimal policy

$$k_{t+1} = \alpha\beta Ak_t^\alpha.$$

(f) Let

$$A = 2, \quad \alpha = 0.35, \quad \beta = 0.95.$$

Use Python to simulate the optimal path of capital for 50 periods starting from $k_0 = 0.5$. Compute the steady state analytically and verify numerically that the simulated path converges toward it.

(g) Repeat the simulation for $\beta = 0.80$ and $\beta = 0.99$. Explain why patience changes both the optimal saving rate and the steady-state capital stock.

Problem 2. A Small Controlled Markov Problem

A student can be in one of three health states:

$$X = \{\text{Sick}, \text{Okay}, \text{Great}\},$$

and can choose either TV or Study.

The reward matrix is

$$R = \begin{pmatrix} 1 & -1 \\ 2 & 1 \\ 8 & 5 \end{pmatrix},$$

where the first column is TV and the second column is Study.

The transition matrix under TV is

$$M_{\text{TV}} = \begin{pmatrix} 0.5 & 0.5 & 0 \\ 0.1 & 0.4 & 0.5 \\ 0 & 0.4 & 0.6 \end{pmatrix},$$

and under Study it is

$$M_{\text{Study}} = \begin{pmatrix} 0.9 & 0.1 & 0 \\ 0.3 & 0.6 & 0.1 \\ 0.3 & 0.6 & 0.1 \end{pmatrix}.$$

Let $\beta = 0.5$.

(a) Write the Bellman equation for each of the three health states.

(b) Start with the policy “TV in every state.” Evaluate this policy by solving

$$V_h = r_h + \beta P_h V_h.$$

(c) Using the value function from part (b), perform one Howard policy improvement step. For each state, calculate the value of choosing TV today and the value of choosing Study today, assuming that continuation values are given by V_h . State the improved policy.

(d) Starting from $V_0 = 0$, implement value function iteration. Iterate until

$$\max_x |V_{n+1}(x) - V_n(x)| < 10^{-10}.$$

Report the converged value function and optimal policy.

(e) Compare the result from Howard’s method with the result from value function iteration. Explain why the methods can follow different numerical paths while solving the same Bellman equation.

(f) Construct the transition matrix implied by the optimal policy and compute its stationary distribution. Interpret the stationary probabilities.

Problem 3. The Advertising Decision

A company is considering whether to make an advertising buy. The buy costs \$1 million.

In a good year, the company has sales of \$4 million; in a normal year, sales are \$2 million; and in a bad year, sales are \$1 million. The company regards an advertising buy as an investment in its future.

If there is *no ad buy*, the transition matrix is

$$M(0) = \begin{pmatrix} 0.8 & 0.2 & 0 \\ 0.2 & 0.8 & 0 \\ 0.1 & 0.4 & 0.5 \end{pmatrix}.$$

If there *is an ad buy*, the transition matrix is

$$M(1) = \begin{pmatrix} 0.5 & 0.5 & 0 \\ 0.1 & 0.6 & 0.3 \\ 0.1 & 0.3 & 0.6 \end{pmatrix}.$$

Rows and columns are ordered

$$\{\text{Bad, Normal, Good}\}.$$

Let

$$\beta = 0.9.$$

Measure all rewards in millions of dollars. Hence the one-period reward matrix is

$$R = \begin{pmatrix} 1 & 0 \\ 2 & 1 \\ 4 & 3 \end{pmatrix},$$

where the first column is No Buy and the second is Buy.

- (1) What are the five elements of a dynamic program as they pertain to this problem?
- (2) Write the Bellman equation when the company is having a normal year.
- (3) Import `numpy`, `pandas`, and `quantecon`. Define the States and Controls as lists. Define the reward as a NumPy array. Set $\beta = 0.9$. Define the value function as an array of zeros, one for each state.
- (4) Make a dictionary called `transition` that has two entries, indexed by 'No Buy' and 'Buy'. The entries are the two Markov matrices.
- (5) Write a function that checks that the value function, reward function, transition rule, and discount factor all make sense. At a minimum, check dimensions, that transition probabilities are nonnegative, that each row of each transition matrix sums to one, and that $0 < \beta < 1$.
- (6) Create a function that takes as inputs the state, the control, and the transition dictionary. It should return the relevant transition probabilities.
- (7) Write a function that updates the value function using the Bellman relation.
- (8) Write a function that iterates on the value function until convergence.

- (9) Write a function that records the optimal policy.
- (10) Write a function that constructs the transition matrix when the company is following the optimal policy. Then compute its ergodic distribution.
- (11) Display the output as a Pandas DataFrame whose rows are indexed by state. The five columns should report:
 - (i) reward if No Buy;
 - (ii) reward if Buy;
 - (iii) solved value function;
 - (iv) optimal policy;
 - (v) ergodic probability of the state.
- (12) Write a class called `DP_Markov` that incorporates everything you have done here.

Problem 4. Economics of the Advertising Policy

Use the advertising model from Problem 3.

- (a) Report the optimal policy in each state and explain the economics of it. Why can it be optimal to advertise in one state but not another even though the advertising cost is always \$1 million?
- (b) Under the optimal policy, compute the stationary distribution of states. Using that distribution, compute long-run average sales and long-run average one-period profit net of advertising expenditure.
- (c) Let the advertising cost be c rather than \$1 million. For each

$$c \in [0, 2]$$

on a fine grid, solve the dynamic program and record the optimal policy.

Create a figure showing how the optimal action in each state changes with c .

- (d) For each value of c , compute the ergodic distribution generated by the optimal policy and the corresponding long-run average sales. Plot long-run average sales against the advertising cost.

Explain why the relationship can have jumps.

- (e) Now return to an advertising cost of \$1 million and vary

$$\beta \in [0.50, 0.99].$$

Determine how the optimal policy changes as the company becomes more patient.

Explain why advertising is fundamentally an intertemporal decision.

- (f) A manager says: “We should advertise whenever the expected increase in next year’s sales exceeds \$1 million.” Explain carefully why this rule does not generally solve the dynamic program.

Problem 5. What Have We Learned?

Write approximately 300–400 words explaining the relationship among the following objects:

state, control, reward, transition rule, value function, policy function.

Your answer should explain why a Bellman equation converts an infinite-horizon problem into a one-period optimization problem with a continuation value.

Then explain the conceptual difference between **value function iteration** and **Howard policy improvement**.

A Final Requirement

At the end of your notebook include a short section entitled

What surprised me?

Identify one result from this assignment that was not obvious to you before doing the calculations. Explain why it surprised you and what you now understand differently.